LegalEase

6. Project Testing Phase

AI-Powered Legal Document Generator

Objective

Verify that LegalEase works end-to-end and that document generation, preview, editing and exports operate correctly.

Functional Tests

- Enter document type, parties, terms and effective date.
- Generate a document through Gemini.
- Confirm generated text appears in the UI.
- Open and edit the generated content.
- Download TXT output.
- Download DOCX output.
- Download PDF output.

Backend Tests

- Verify the FastAPI service starts successfully with Uvicorn.
- Verify the root endpoint responds.
- Verify POST /generate accepts the required request fields.
- Verify Pydantic validation handles incoming data.
- Verify GeminiDocumentGenerator is called by the route.

Output Validation

- Check that DOCX contains the expected title, formatting, logo, terms table and footer.
- Check that PDF contains the logo and footer on pages as specified.
- Check bullet-style terms and clean formatting.
- Check that TXT contains the generated editable text.

Integration Testing

- Test the full path: Streamlit → FastAPI → Gemini → response → preview → edit → export.
- Test different document types and combinations of parties, terms and dates.

- Test Gemini integration under different input combinations.

Deployment Verification

- Install dependencies using requirements.txt.
- Configure the Gemini API key in .env.
- Launch FastAPI with Uvicorn.
- Launch the Streamlit app.
- Verify all major features in the local environment.